1. Prove by induction that if Euclid's algorithm requires N divisions to compute the greatest common divisor of the pair $a > b, a, b \in \mathbb{N}$, then $a \geq F_{N+2}$ and $b \geq F_{N+1}$, where F_i denotes the ith Fibonacci number ($F_1 = 1, F_2 = 1, F_{i+2} = F_{i+1} + F_i$).

1) for $N=1$, $a = rb$. $a, b \in \mathbb{N}$ and $a > b$ so $b \geq 1 = f_2$ and $a \geq 2 = f_3$.
Assume statement to be true for $n = k$ so if we require N divisions then $a \geq f_{N+2}$ and $b \geq f_{N+1}$.
If $N+1$ divisions are required for (α, a) then $\alpha = ra + b \geq a + b$ as $r \geq 1$ and since $a \geq f_{N+2}$, $b \geq f_{N+1}$ we have $\alpha \geq f_{N+2} + f_{N+1}$ and hence $\alpha \geq f_{N+3}$. Therefore, we have $\alpha \geq f_{(N+2)+1}$ and $a \geq f_{(N+1)+1}$ for $N+1$ divisions and so by induction we $a \geq f_{N+2}$ and $b \geq f_{N+1}$ $\forall N \in \mathbb{N}$.